

Civ Eng 2200 Statics
Spring 2022 Exam 3 (for practice)

Section 1 – Internal Forces

1. (5 pt) What approach is typically applied when determining the internal loads within a rigid body?

- A. method of sections
- B. method of joints
- C. integration
- D. two-force member analysis
- E. particle equilibrium analysis

2. (5 pt) The force that acts perpendicular to the “sectioned” surface is called the _____.

- A. normal force
- B. bending force
- C. universal force of attraction
- D. shear force
- E. gravitational force
- F. centrifugal force

3. (5 pt) Determine the magnitude of the shear force at section $b-b$ that lies 0.1 m from A, located 0.1 m along crank arm OA, if $m = 5$ kg.

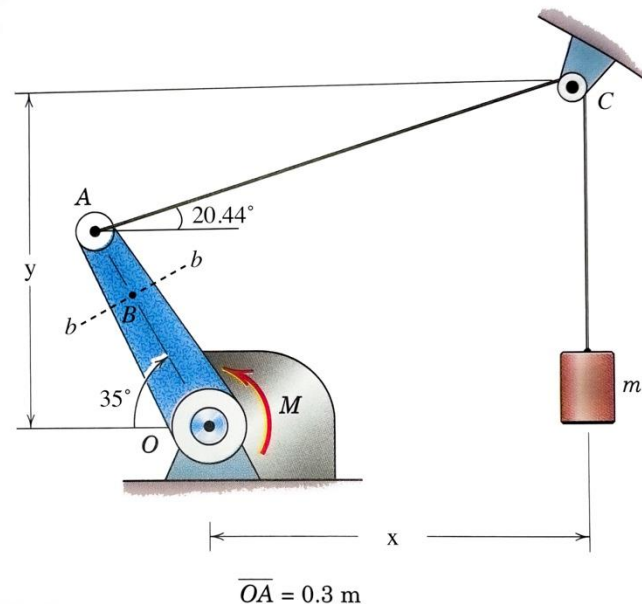
- A. 40.4 N
- B. 23.5 N
- C. 34.1 N
- D. 68.2 N
- E. 52.7 N

4. (5 pt) Determine the magnitude of the normal force at section $b-b$ that lies 0.1 m from A, located 0.1 m along crank arm OA, if $m = 5$ kg.

- A. 27.8 N
- B. 13.3 N
- C. 32.0 N
- D. 21.4 N
- E. 19.5 N

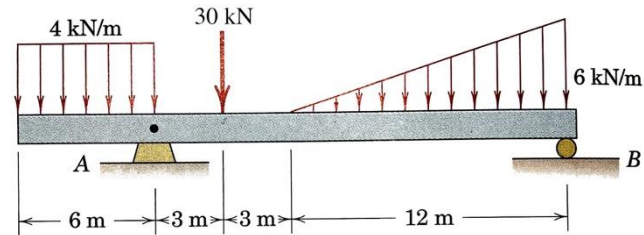
5. (5 pt) Determine the magnitude of the bending moment at section $b-b$ that lies 0.1 m from A, located 0.1 m along crank arm OA, if $m = 5$ kg.

- A. 4.04 N-m
- B. 4.79 N-m
- C. 3.26 N-m
- D. 3.01 N-m
- E. 2.38 N-m



Section 2 – Shear and Moment Diagrams

Provide information for constructing the **shear diagram**. The x origin is on the left and the support reactions are $A_y = 61$ kN acting up, $B_y = 29$ kN acting up. A is a pin and B is a roller.



6. (3 pt) What happens to the shear diagram at $x = 6$ m?

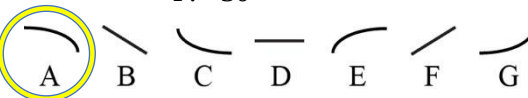
- | | |
|-------------------|-------------------|
| A. Increase 61 kN | D. Decrease 45 kN |
| B. Decrease 24 kN | E. Increase 45 kN |
| C. Increase 24 kN | F. Decrease 30 kN |

Determine the numerical value (magnitude only) of the change in shear for section $12 < x < 24$ m as well as the behavior (shape) of the shear for the same region.

7. (3 pt) Change = _____ kN (magnitude only)

- | | |
|-------|-------|
| A. 36 | D. 24 |
| B. 29 | E. 54 |
| C. 61 | F. 30 |

8. (3 pt) Behavior:



9. (3 pt) What happens to the shear diagram at $x = 9$ m?

- | | |
|-------------------|-------------------|
| A. Decrease 30 kN | D. Increase 54 kN |
| B. Increase 24 kN | E. Decrease 54 kN |
| C. Decrease 24 kN | F. Decrease 36 kN |

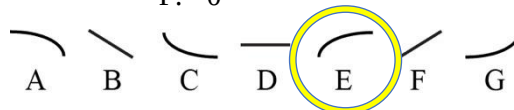
Provide information for constructing the **moment diagram**. The x origin is on the left and the support reactions are $A_y = 945.455$ N acting up, $B_y = 2854.55$ N acting up. A is a pin and B is a roller.

Determine the numerical value (magnitude only) of the change in moment for section $2 < x < 5.113$ m as well as the behavior (shape) for the same region.

10. (3 pt) Change = _____ N-m (magnitude only)

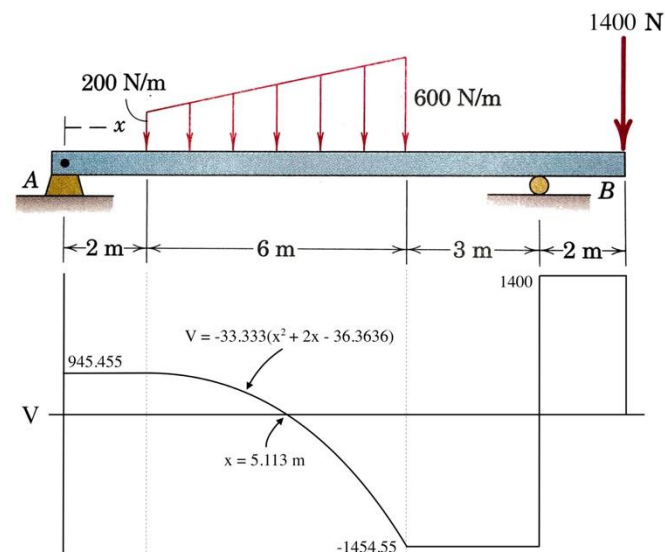
- | | |
|-----------|-----------|
| A. 1638.9 | D. 1966.2 |
| B. 1890.9 | E. 4363.6 |
| C. 945.45 | F. 0 |

11. (3 pt) Behavior:



12. (3 pt) What happens to the moment diagram at $x = 11$ m?

- | | |
|---------------------|----------------------|
| A. No change | D. Decrease 400 N-m |
| B. Decrease 200 N-m | E. Increase 400 N-m |
| C. Increase 600 N-m | F. Decrease 2400 N-m |



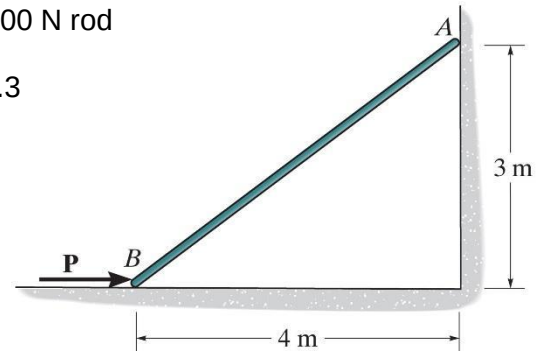
13. (4 pt) Determine the order of the equation for the internal bending moment for the region $8 < x < 11$ m on the beam shown.

A. first order, linear
B. constant
C. second order, parabolic
D. zero
E. third order, cubic

Section 3 - Friction

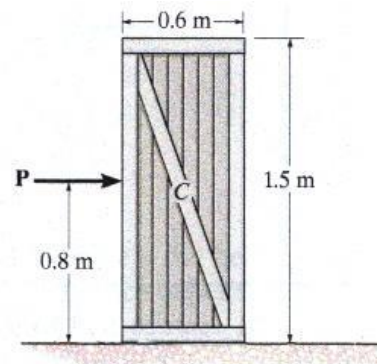
14. Determine the minimum force **P** to initiate motion of the uniform 300 N rod **AB** up the wall. The contact surface at **A** is smooth, whereas the coefficient of static friction between the rod and the floor at **B** is $\mu_s = 0.3$

A. 90 N
B. 190 N
C. 290 N
D. 300 N
E. 390 N



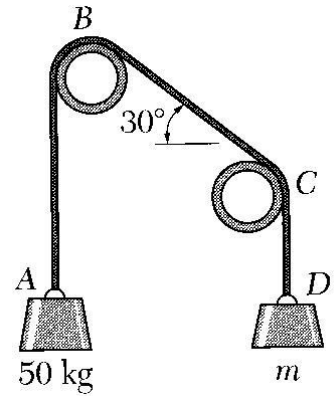
15. Determine the minimum horizontal force **P** required to cause the crate to move. The crate weighs 400 N and the coefficient of static friction between the crate and the plane is $\mu_s = 0.4$.

A. 100 N
B. 120 N
C. 130 N
D. 150 N
E. 180 N



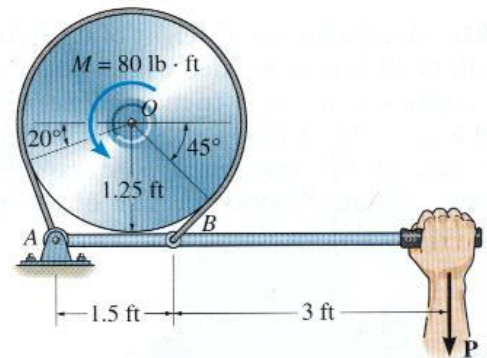
16. The coefficient of friction is $\mu = 0.15$ between the fixed posts at B and C and the cord. What is the mass m of D required so that motion of D impends downward?

- A. 60 kg
- B. 70 kg
- C. 80 kg**
- D. 90 kg
- E. 100 kg



17. Force P is applied to the handle and belt assembly to prevent motion of the drum. If motion of the drum impends when an 80 lb-ft torque is applied to the drum as shown, which cable is T_2 ?

- A. Cable A
- B. Cable B**

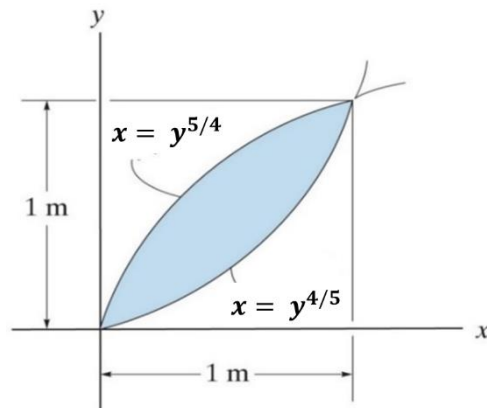


18. To test if a wedge is self-locking, one does the following:
- A. Apply force P pushing on the wedge with $P = 0$.
 - B. Apply force P pulling on the wedge with $P = 0$.**
 - C. Apply force P on the wedge with $P = 0$ for the raising case.
 - D. Apply force P pushing on the wedge with $\mu = 0$.
 - E. Apply force P pulling on the wedge with $\mu = 0.5$.

Section 4 – Centroids

19. (5 pts) When using a **horizontal** differential element area to solve for the location of $\bar{x}$ for the shaded area, what expression would be correct for dA ?

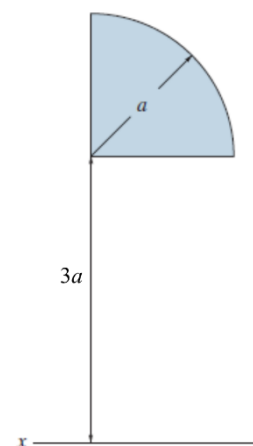
- A. $(y^{4/5} - y^{5/4}) dy$
- B. $(y^{4/5} + y^{5/4}) dy$
- C. $0.5(y^{5/4} - y^{4/5}) dy$
- D. $(y^{5/4} - y^{4/5}) dy$
- E. $0.5(y^{4/5} + y^{5/4}) dy$



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20. (5 pts) A ring is generated by rotating the quarter-circular area about the x axis. Determine its volume if the radius $a = 2$ cm.

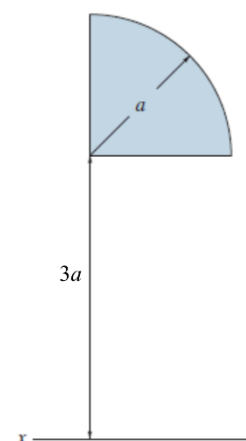
- A. 135.2 cm^3
- B. 125.7 cm^3
- C. 153.2 cm^3
- D. 140.1 cm^3
- E. 177.6 cm^3



21. (5 pts) A ring is generated by rotating the quarter-circular area about x axis. Determine its surface area if the radius $a = 2$ cm.

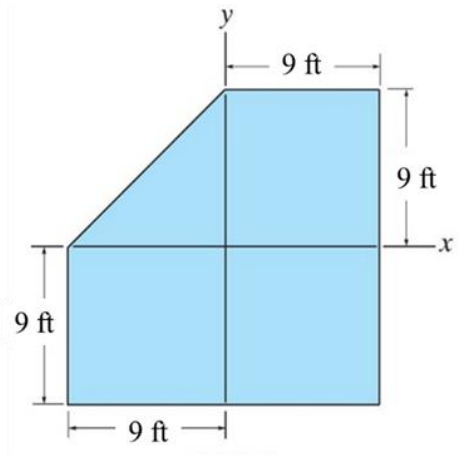
- A. 319.5 cm^2
- B. 306.9 cm^2
- C. 297.5 cm^2
- D. 389.7 cm^2
- E. 200.1 cm^2

the



22. (5 pt) Determine the location of $\bar{y}$ for the shaded area.

- A. -0.86 ft
- B. -0.57 ft
- C. -0.75 ft
- D. -0.91 ft
- E. -0.68 ft



23. (5 pt) Determine the location of $\bar{z}$ for the composite rod if $h = 6$ ft.

- A. 0.97 ft
- B. 0.86 ft
- C. 0.91 ft
- D. 0.78 ft
- E. 0.62 ft

